LDA MODEL

Code

Hide

```
library(readr)     # for reading CSV
library(MASS)      # for lda()
library(ggplot2)   # for visualization
```

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```
file.choose()
```

```
[1] "C:\\Users\\SuzLiz\\OneDrive\\SEM 6\\DSP 2\\GP PROJ\\wheelmtbf.csv"
```

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```
# Read the CSV file
data <- read_csv("C:\\Users\\SuzLiz\\OneDrive\\SEM 6\\DSP 2\\GP PROJ\\wheelmtbf.csv")
```

```
Rows: 4923 Columns: 12── Column specification ──────────────────────────────────────────────
────────────────────────────────────────────────────
Delimiter: ","
chr  (5): TrainID, Car, Bogie, Axle, Quarter
dbl  (6): WheelDiameter, WheelFlange, Failure, FailuresPerQuarter, DaysInQuarter, MTBF
date (1): Date
ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

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```
data
```

Date <date>	TrainID <chr>	C.. <chr>	Bo... <chr>	Axle <chr>	WheelDiameter <dbl>	WheelFlange <dbl>	Failure <dbl>	Quarter <chr>	Failu
2018-06-22	T01	M1	BOG#1	X1-LHS	843.22	29.02	0	2018Q2	
2018-06-22	T01	M1	BOG#1	X1-RHS	843.93	29.26	0	2018Q2	
2018-06-22	T01	M1	BOG#1	X2-LHS	844.14	29.80	0	2018Q2	
2018-06-22	T01	M1	BOG#1	X2-RHS	844.76	29.53	0	2018Q2	
2018-06-22	T01	M1	BOG#2	X3-LHS	844.86	29.80	0	2018Q2	
2018-06-22	T01	M1	BOG#2	X3-RHS	845.75	29.77	0	2018Q2	

Date	TrainID	C..	Bo...	Axle	WheelDiameter	WheelFlange	Failure	Quarter	Failu
<date>	<chr>	<chr>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<chr>	
2018-06-22	T01	M1	BOG#2	X4-LHS	843.62	29.78	0	2018Q2	
2018-06-22	T01	M1	BOG#2	X4-RHS	844.19	29.60	0	2018Q2	
2018-06-22	T01	T1	BOG#1	X1-LHS	843.25	30.54	0	2018Q2	
2018-06-22	T01	T1	BOG#1	X1-RHS	844.23	30.06	0	2018Q2	

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```
# Convert 'FailurePerQuarter' to a factor (for classification)
data$Failure <- as.factor(data$Failure)
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```
# Fit the Linear Discriminant Analysis model
lda_model <- lda(Failure ~ WheelDiameter + WheelFlange, data = data)
```

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```
# Display LDA summary
print(lda_model)
```

```
Call:
lda(Failure ~ WheelDiameter + WheelFlange, data = data)

Prior probabilities of groups:
        0         1 
0.8630916 0.1369084 

Group means:
  WheelDiameter WheelFlange
0      812.9757    29.60291
1      798.7814    26.73681

Coefficients of linear discriminants:
                       LD1
WheelDiameter  0.002629184
WheelFlange   -0.898452568
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```
# Make predictions using the LDA model
lda_pred <- predict(lda_model)
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```
# Add predicted class to the cleaned dataset
data$Predicted <- lda_pred$class
```

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```
# Create scatter plot with classification results
ggplot(data, aes(x = WheelDiameter, y = WheelFlange, color = Predicted, shape = Failure)) +
  geom_point(size = 3) +
  labs(title = "LDA Separation and Classification",
       x = "Wheel Diameter", y = "Wheel Flange") +
  theme_minimal()
```